CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CL)

May 2012

CL208 Advanced Surveying

Date: 15.05.2012, Tuesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain procedure to determine additive and multiplying constant for tacheometer. **[04]**

(b) Define **[03]**

1. Observed value of quantity 2. Most probable value of quantity
3. True value of quantity.

Q - 2 (a) A tacheometer is setup at an intermediate point on a traverse course PQ and the following observations are made on a vertically held staff. **[05]**

Staff Station	Vertical angle	Staff intercept	Axial hair Reading
P	+9° 30'	2.250	2.105
Q	+6° 00'	2.055	1.875

The instrument is fitted with anallactic lens and the multiplying constant is 100. Compute the length PQ and the reduce level of Q. Take R.L. of P = 350.50m.

(b) Adjust the angle P, Q, R and S which close the horizon. **[05]**

$\angle P = 100° 30' 22''$ Weight 1 $\angle Q = 80°40'10''$ Weight 2

$\angle R = 90°20'8''$ Weight 3 $\angle S = 88°29'25''$ Weight 4

(c) Differentiate raster data structure and vector data structure in GIS. **[04]**

OR

Q – 2 (a) What is tangential method of tacheometry? Derive elevation and distance formula for tangential method when one angle is the angle of elevation and the other is angle of depression. **[05]**

(b) Angles were measure on a station and the observations were recorded as follows: **[05]**

$\angle A = 45°30'10''$ Weight 2 $\angle B = 40°20'20''$ Weight 3

$\angle A+B = 85°50'10''$ Weight 1

Find the most probable vale of angle A and B.

(c) What do you understand by GPS? What are its main advantages over traditional method of surveying? **[04]**

Q – 3 (a) Discuss the subtense bar method of tacheometric surveying. What are its advantages and disadvantages over the stadia method? **[04]**

(b) Explain laws of weight in detail. **[04]**

(c) List the application of remote sensing and describe any two in details. **[06]**

OR

Q – 3 (a) Derive elevation and distance formula for inclined sight with staff normal to line of sight. [04]

(b) Enlist types of errors and explain any one in detail. [04]

(c) Differentiate active system and passive system in remote sensing. [06]

SECTION – II

Q - 4 (a) Define the followings: [05]

(i) Scale (ii) Tilt & Tip (iii) Iscocentre (iv) Principal point (v) Floating marks

(b) (i) Convert the angle in hours, minutes and seconds 5°15'25''. [02]

(ii) Define phase of signal

Q - 5 (a) Determine the R.L. of the top of a transmission tower from the following observations. [07]

Inst. Station	Vertical angle top of tower	Staff reading on B.M	RL. of B.M
A	18°30'	2.815m	105.00m
B	12°40'	1.865m	

Also determine distance between the station A and the transmission tower. The distance between the station A and B is 60m. The station A, B and the tower are in the same vertical plane.

(b) Explain scale of vertical and tilted photographs in detail. [07]

OR

Q - 5 (a) In order to determine the elevation of the top Q of a signal on a hill, observations were made from two stations P and R. The stations P, R and the signal Q were in the same plane. If the angles of elevation of the top Q of the signal measured at P and R were 25°35' and 15°5' respectively, determine the elevation of the foot of the signal if the height of the signal above it base was 4m. The staff readings upon the bench mark (R.L. 105.42) were respectively 2.755m and 3.855m, when the instrument was at P and at R. The distance between P and R was 120m. [07]

(b) Write short note on Parallax. [07]

Q - 6 (a) What do you understand by horizontal control? What are different methods of establishing a horizontal control? [07]

(b) Find the azimuth and altitude of a star from the following data. [07]

Latitude of the place = 46° N

Hour angle of the star = $20^h 40^m$

Declination =18°38'S

OR

Q - 6 (a) What are the uses of triangulation survey? [07]

(b) Calculate the hour angle and azimuth of the sun at sunrise when its declination is 18°N. The latitude of the place is 46°S. [07]

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